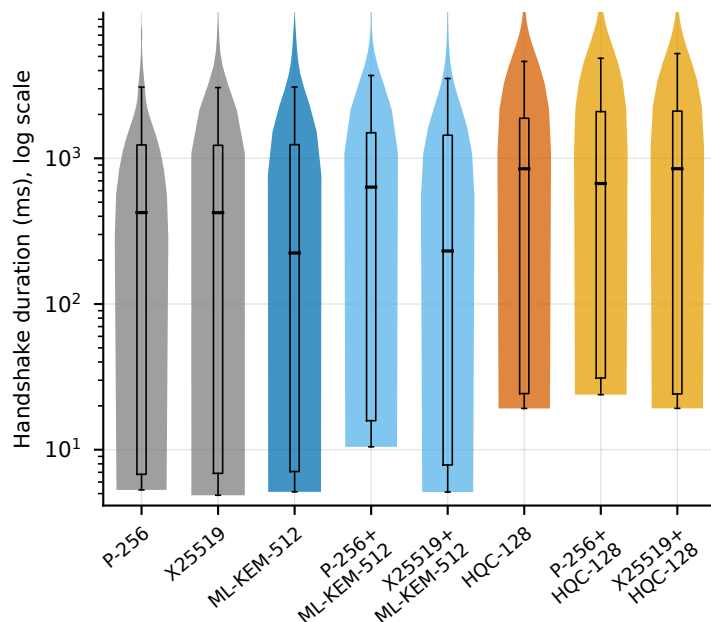
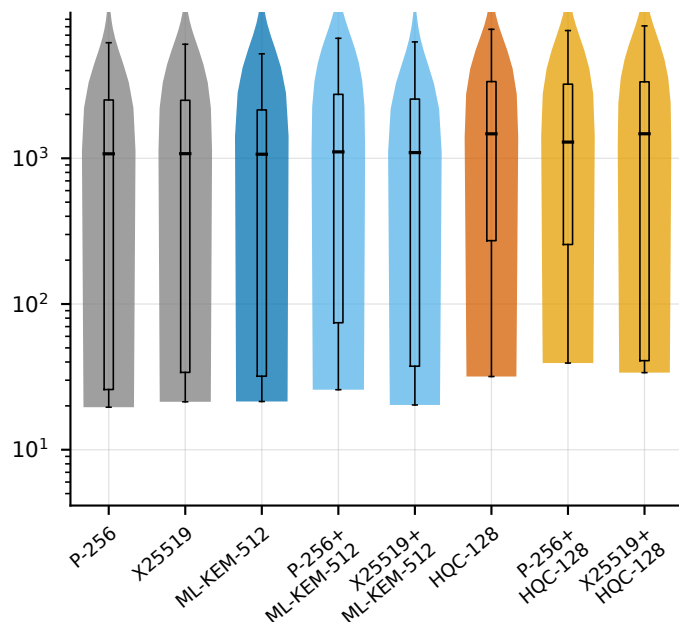


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	1.07k	191.0	27.63k	0	3.09k	293.7	109.08k	0
X25519	1.44k	377.0	106.30k	1/500	3.35k	339.0	108.53k	3/500
ML-KEM-512	1.57k	328.0	75.13k	1/500	3.33k	350.0	138.14k	3/500
P-256+ML-KEM-512	1.66k	347.4	106.00k	1/500	4.14k	327.2	111.21k	1/500
X25519+ML-KEM-512	1.49k	279.9	53.34k	0	3.00k	265.6	107.01k	4/500
HQC-128	2.87k	343.7	107.43k	2/500	4.20k	283.3	134.88k	4/500
P-256+HQC-128	3.14k	374.2	108.53k	1/500	4.00k	312.7	135.66k	3/500
X25519+HQC-128	2.96k	326.9	112.93k	3/500	4.64k	283.9	117.06k	4/500